Clinical Genomics Evidence & Deep Research Synthesis: Melinda Ehrle

Patient / Cohort Identifier: `Melinda_Ehrle` | **Report Date:** October 13, 2026

Target Audience: Medical Geneticist · Subspecialty Investigator · Primary Care Physician · Clinical Bioinformatician

Sequencing Depth & Reference: BAM Aligned Mean Depth **33.4x** on GRCh38.p14

Orientation: What We Are Covering

This clinical genomics research dossier evaluates **1653 prioritized variants across 1129 clinical genes** and **41 pharmacogenes (377 screened medications)** for **Melinda Ehrle** (`Melinda_Ehrle`), backed by **33.4x** BAM mean sequencing depth. This synthesis separates **Mechanistic Causality** (coding loss-of-function, canonical splice disruption, 3D active-site destabilization) from **Statistical Correlation** (GWAS linkage disequilibrium proxies and common polygenic modifiers), audits **Intra-Gene Cis/Trans Phasing** across multi-variant genes, and provides a **Biomarker Ground-Truthing & Longitudinal Futures Roadmap**.

Body

1. Primary Mechanistic & High-Priority Clinical Findings (With Locus Depth, VAF & Confidence)

Gene	Variant & Coordinate (GRCh38)	Zygosity & Phase	Locus DP & VAF	ClinVar Status	CADD (Phred) / REVEL	AlphaGenome AVI (Phred) & Top Driver	Confidence
POLG	`p.Gly737Arg` (`chr15:89323460`)	Heterozygous (Paternal)	`31x` / `0.61`	**Pathogenic/Likely pathogenic**	26.3 / 0.94	30.3 (AlphaMissense)	**0.95**
TAT	`p.Arg57Ter` (`chr16:71576247`)	Heterozygous (Maternal)	`52x` / `0.46`	**Pathogenic**	36.0 / —	38.8 (Protein Termination)	**0.95**
ATM	`c.1236-2del` (`chr11:108250699`)	Heterozygous (Unknown)	`18x` / `0.17`	**Pathogenic/Likely pathogenic**	— / —	—	**0.71**
VDR	`c.-1172A>G` (`chr12:47906043`)	Heterozygous (Unphased)	`21x` / `0.19`	**Likely pathogenic**	5.4 / —	2.1 (DNASE-seq)	**0.68**
MC1R	`p.Asp294His` (`chr16:89920138`)	Heterozygous (Paternal)	`25x` / `0.52`	**Conflicting classifications of pathogenicity**	risk factor	26.9 / 0.61	24.5 (Cactus)	**0.82**
AMPD1	`p.Glu435Gly` (`chr1:114677435`)	Heterozygous (Unphased)	`41x` / `0.49`	**Uncertain significance**	32.0 / 0.95	28.2 (AlphaMissense)	**0.88**
CTH	`p.Thr67Ile` (`chr1:70415987`)	Heterozygous (Unphased)	`39x` / `0.46`	**Conflicting classifications of pathogenicity**	25.5 / 0.93	27.8 (AlphaMissense)	**0.88**

2. Mechanistic Causality vs. Statistical Correlation Matrix

To prevent clinical over-interpretation of statistical GWAS proxies, cohort findings are stratified across four epistemic tiers:

Epistemic Class	Count in Cohort	Representative Genes	Mechanistic vs. Statistical Distinction & Clinical Rule
Q1 · Confirmed Mechanistic Causality	**5**	ATM, POLG, TAT, MC1R, CYP3A5	**Direct Physical Disruption + Clinical Proof:** Concordant ClinVar P/LP curation backed by high AI functional impact (CADD ≥ 20 / AVI ≥ 20). Eligible for preventive clinical surveillance.
Q2 · Cryptic AI-Predicted Disruptors	**159**	HLA-DRB5, CEP63, GUSB, HLA-DRB1, IL17RC, TEK	**Biophysical Disruption Without Clinical Database Entry:** Severe deep-learning predicted splice, chromatin, or 3D folding impact currently VUS/unreviewed in ClinVar. Prioritize for biomarker ground-truthing.
Q3 · Database-Curated / Statistical & PGx Modifiers	**377**	CD40, HEPACAM, HLA-DRB1, TAT, VPS35L, VDR	**Curated Hypomorphic, PGx & Statistical GWAS Loci:** Includes enzymatic drug-response alleles and statistical population modifiers; must **never** trigger invasive monogenic workups.
Q4 · Protective & Resilience Offsets	**10**	ADH1C, GNPTAB, TLR1, CASP8, EDNRA, FCGR2B	**Risk-Attenuating & Cardioprotective Alleles:** Curated protective modifiers buffering baseline metabolic or vascular pathway susceptibility.

2.1 High-Priority Cryptic AI Disruptors (AlphaGenome 18-Modality Attribution)

Gene	Variant & Coordinate	Zygosity & Phase	AVI (Phred) & Percentile	Primary Modality Driver	Top SHAP Attribution Weights (`fi_*`)
VWA3B	`p.Glu372Ter` (`chr2:98162976`)	Heterozygous (Unphased)	**50.8** (Top 0.00%)	**Protein Termination**	Protein Term: 52.0%, Splicing: 18.9%, Cactus 241-Way: 10.2%
CEP63	`p.Gln185His` (`chr3:134537268`)	Heterozygous (Maternal)	**38.6** (Top 0.01%)	**Splicing**	Splicing: 40.3%, Cactus 241-Way: 23.1%, AlphaMissense: 19.6%
OGFOD3	`c.380+1G>A` (`chr17:82411454`)	Heterozygous (Unphased)	**33.8** (Top 0.04%)	**Splicing**	Splicing: 53.6%, Cactus 241-Way: 27.2%, PhastCons 470-Way: 8.6%
NIPSNAP2	`c.712+1G>T` (`chr7:55994989`)	Heterozygous (Unphased)	**33.8** (Top 0.04%)	**Splicing**	Splicing: 58.6%, Cactus 241-Way: 27.4%, PhastCons 470-Way: 9.4%
HLA-DRB5	`c.101-1G>A` (`chr6:32522175`)	Heterozygous (Unphased)	**28.0** (Top 0.16%)	**Splicing**	Splicing: 76.6%, PhastCons 470-Way: 12.7%, Cactus 241-Way: 4.5%
AMPD1	`p.Glu435Gly` (`chr1:114677435`)	Heterozygous (Unphased)	**28.2** (Top 0.15%)	**AlphaMissense**	AlphaMissense: 51.3%, Cactus 241-Way: 25.4%, PhastCons 470-Way: 14.7%

3. Intra-Gene Cis / Trans Phase Auditor & Cross-Pathway Epistasis

Parental and read-backed phasing across all multi-variant genes in **Melinda Ehrle** resolved **3 Trans Multi-Variant Genes** (variants on opposite parental homologs; 0 confirmed biallelic pathogenic recessive compound-heterozygotes) and **7 Cis Haplotype Genes** (variants co-inherited on the same parental chromosome, leaving the opposite parental homolog completely wild-type).

Gene	Organ System	Phased Architecture	Variant Count (Mat / Pat)	Max CADD (Phred)	Clinical & Inheritance Interpretation
DNAH11	Heart & Cardiovascular	**TRANS (Biallelic / Compound Het Candidate)**	4 (1M / 1P)	21.8	1 Maternal + 1 Paternal alleles in trans across DNAH11
ABCB1	Multisystem	**TRANS (Biallelic / Compound Het Candidate)**	17 (1M / 7P)	7.0	1 Maternal + 7 Paternal alleles in trans across ABCB1
UGT2B7	Multisystem	**TRANS (Biallelic / Compound Het Candidate)**	14 (1M / 1P)	6.7	1 Maternal + 1 Paternal alleles in trans across UGT2B7
TMEM43	Heart & Cardiovascular	**CIS (Paternal Co-Inherited Haplotype)**	2 (0M / 2P)	32.0	2 variants co-inherited in cis on Paternal chromosome (WT Maternal copy intact)
PTPRC	Multisystem	**CIS (Maternal Co-Inherited Haplotype)**	3 (2M / 0P)	22.8	2 variants co-inherited in cis on Maternal chromosome (WT Paternal copy intact)
ATR	Multisystem	**CIS (Paternal Co-Inherited Haplotype)**	2 (0M / 2P)	17.3	2 variants co-inherited in cis on Paternal chromosome (WT Maternal copy intact)
PCSK9	Heart & Cardiovascular	**CIS (Paternal Co-Inherited Haplotype)**	8 (0M / 2P)	14.3	2 variants co-inherited in cis on Paternal chromosome (WT Maternal copy intact)
RYR1	Heart & Cardiovascular	**CIS (Paternal Co-Inherited Haplotype)**	15 (0M / 11P)	10.4	11 variants co-inherited in cis on Paternal chromosome (WT Maternal copy intact)
APOE	Multisystem	**CIS (Maternal Co-Inherited Haplotype)**	3 (2M / 0P)	10.2	2 variants co-inherited in cis on Maternal chromosome (WT Paternal copy intact)
ETNK2	Multisystem	**CIS (Paternal Co-Inherited Haplotype)**	3 (0M / 2P)	1.6	2 variants co-inherited in cis on Paternal chromosome (WT Maternal copy intact)

4. Actionable Pharmacogenomic (CPIC / DPWG) Medication Matrix

Prescribing Action	Screened Drug Count	Key Affected Medications & Causal Pharmacogenes
🚫 **Avoid / Contraindicated**	**1**	**Clopidogrel (Plavix)** (CYP2C19)
⬇️ **Reduce Dose**	**1**	**Warfarin (Coumadin)** (VKORC1)
⚠️ **Use With Caution**	**9**	**Antineoplastic Agents** (ABCG2), **Apixaban** (ABCG2), **Dolutegravir** (ABCG2), **Felodipine** (ABCG2), **Iguratimod** (ABCG2), **Sorafenib** (ABCG2), **Sunitinib** (ABCG2), **Tenofovir** (ABCG2)
🔍 **Monitor**	**1**	**Irinotecan** (RGS5/RXRG/NR1I2/ABCG2)
✅ **Preferred Alternatives**	**55**	**Beta Blocking Agents, Selective** (), **Bisphosphonates** (FDPS), **Opioid Anesthetics** (ABCG2), **Other General Anesthetics** (ABCG2), **Vitamin B-Complex, Incl. Combinations** (MTHFR), **Amisulpride** (), **Atenolol** (MOV10/AGT/ADD1/GRK4/SLC25A31/ADRB2), **Atorvastatin** (PRDM16/AGT/SCAP/ABCG2/MYLIP)

4.1 Allelic Fraction (VAF) Skewing & Mosaicism / Paralog Uncertainty Audit

Gene	Variant & Coordinate	Locus DP	Observed VAF	Classification	Technical Interpretation
ATM	`c.1236-2del` (`chr11:108250699`)	`18x`	**0.17**	Pathogenic/Likely pathogenic	Skewed VAF (<0.35); rule out clonal mosaicism vs. paralog alignment skew before clinical action.
VDR	`c.-1172A>G` (`chr12:47906043`)	`21x`	**0.19**	Likely pathogenic	Skewed VAF (<0.35); rule out clonal mosaicism vs. paralog alignment skew before clinical action.
SLC27A3	`p.Leu173Pro` (`chr1:153776015`)	`13x`	**0.23**	Not reviewed	Skewed VAF (<0.35); rule out clonal mosaicism vs. paralog alignment skew before clinical action.
UCK2	`c.99+808G>T` (`chr1:165828740`)	`28x`	**0.32**	Not reviewed	Skewed VAF (<0.35); rule out clonal mosaicism vs. paralog alignment skew before clinical action.
SYNPR	`c.84+40453T>G` (`chr3:63319195`)	`17x`	**0.29**	Not reviewed	Skewed VAF (<0.35); rule out clonal mosaicism vs. paralog alignment skew before clinical action.
SMPD4	`p.Pro217Leu` (`chr2:130172358`)	`21x`	**0.33**	Not reviewed	Skewed VAF (<0.35); rule out clonal mosaicism vs. paralog alignment skew before clinical action.

Conclusions: Diagnostic & Epistemic Decision Calculus

Arguments FOR Preventive Surveillance & Genotype-Guided Prescribing

1. **High-Depth Analytical Grounding:** Empirical BAM aligned coverage (**33.4x** mean depth) provides high analytical certainty (`0.98`) across SNVs and small indels.

2. **Actionable Pharmacogenomic Safety:** 12 drug-gene interactions across 41 pharmacogenes provide immediate CPIC/DPWG guidance to avoid toxicity and optimize dosing.

3. **Low-Cost Biomarker Ground-Truthing:** Routine PCP blood panels directly verify physiological homeostasis across the sample's specific prioritized pathways.

Arguments AGAINST Over-Intervention (Why Invasive Testing Is Unwarranted)

1. **Heterozygous Recessive Carrier & Modifier Reserve:** Heterozygous variants affecting or potentially affecting one allele in autosomal recessive or modifier genes and uni-allelic *Cis* haplotypes (7 genes) retain one intact wild-type parental copy; invasive diagnostic procedures or restrictive diets are clinically unwarranted.

2. **Statistical GWAS ≠ Monogenic Causality:** Common polygenic modifiers (AF > 1%) reflect population linkage disequilibrium proxies and must never trigger monogenic disease interventions.

Opportunities: Physiological Ground-Truthing & Longitudinal Futures Roadmap

1. **Primary Care Preventive Blood Panel:** Incorporate the sample-specific venous blood panels and preventive evaluations from Section 2 of the Clinical Brief into the annual PCP physical.

2. **Cryptic Splice & Regulatory Functional Validation (Quadrant 2):** The **159 Cryptic AI Disruptors** exhibiting high AlphaGenome `AVI` scores driven by splicing or RNA-seq modalities can be inspected via the AlphaGenome Ref vs. Alt Track Predictions view.

3. **Automated Annual Re-Annotation & SHA-256 State Fingerprint:** Verified 0 variant reclassifications since prior snapshot (2026-10-13 11:57:25; SHA-256 c49df28612de0eff).

Confidence & Uncertainty Assessment

- **Analytical WGS Callset & Phasing Confidence: 0.98** (BAM Mean Depth `33.4x`).
- **CPIC / DPWG Pharmacogenomic Prescribing Confidence: 0.95** (41 pharmacogenes, 377 screened drugs).
- **Monogenic Carrier & Surveillance Confidence: 0.94** (Orthogonal ClinVar, OMIM, CADD, and AlphaGenome agreement).
- **Polygenic / VUS Modifier Penetrance Confidence: 0.70** (Requires laboratory biomarker ground-truthing).

Appendix: Supporting Documentation & Variant Cross-Reference Directory

Gene	Variant	Coordinates (GRCh38)	ClinVar	dbSNP	OMIM	ClinGen	AlphaGenome Atlas
POLG	`p.Gly737Arg`	`chr15:89323460`	VCV13513	rs121918054	MIM:203700	POLG	Atlas
TAT	`p.Arg57Ter`	`chr16:71576247`	VCV402	rs118203914	MIM:276600	TAT	Atlas
ATM	`c.1236-2del`	`chr11:108250699`	VCV574589	rs1565381646	—	ATM	Atlas
VDR	`c.-1172A>G`	`chr12:47906043`	VCV3336650	rs4516035	—	VDR	Atlas
MC1R	`p.Asp294His`	`chr16:89920138`	VCV14307	rs1805009	MIM:203200	MC1R	Atlas
AMPD1	`p.Glu435Gly`	`chr1:114677435`	VCV2173226	rs764950035	—	AMPD1	Atlas
CTH	`p.Thr67Ile`	`chr1:70415987`	VCV2939	rs28941785	MIM:219500	CTH	Atlas